

# HEMANTH KUMAR N

Data Engineer | Azure | Microsoft Fabric | PySpark | Databricks | ADF | ADLS Gen2

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## PROFESSIONAL SUMMARY

Results-driven **Data Engineer** with 3 years of experience building reliable, scalable pipelines on **Azure** and **Microsoft Fabric**. Delivered production-grade ETL/ELT systems for Microsoft-scale workloads using **Azure Data Factory**, **Databricks**, **PySpark**, and **Fabric Data Engineering** - cut query runtimes by up to 40% and enabled analytics teams to meet SLAs consistently. Track record of owning solutions end-to-end, from stakeholder requirements to live monitoring with **Azure Monitor**.

## TECHNICAL SKILLS

### Cloud Platform:

Microsoft Azure | Azure Data Factory | ADLS Gen2 | Databricks | Azure Monitor | Azure Synapse | Microsoft Fabric | Fabric Data Factory | Lakehouse | OneLake | Fabric Data Engineering | Azure-Fabric Migration

**Data Engineering:** ETL/ELT Design, Data Modeling, Partitioning & Indexing, Performance Tuning

**Big Data:** Apache Spark, **PySpark**, **Databricks**, Pandas

**Programming & Query:** Python, SQL

**Orchestration & Monitoring:** **Azure Data Factory** Pipelines, **Microsoft Fabric** Data Pipelines, **Azure Monitor**, Azure Log Analytics

## PROFESSIONAL EXPERIENCE

### Data Engineer | Infosys (Microsoft Client) - Bengaluru

April 2024 - Present

- Engineered 10+ **ADF** + **Databricks** ETL pipelines ingesting structured and semi-structured data into **ADLS Gen2**, reducing data latency from source to reporting layer by ~35%.
- Authored **PySpark** transformation modules on **Databricks** to process 50M+ records per run, enabling analysts to query refreshed datasets within scheduled SLA windows - zero breaches over 12 months.
- Redesigned table schemas using partition pruning and composite indexing strategies, cutting average query execution time by 40% across critical analytics workloads.
- Delivered **SQL** performance tuning across 15+ stored procedures and views, eliminating full table scans and reducing report generation time from 8 minutes to under 2 minutes.
- Established pipeline observability using **Azure Monitor** alerts and structured logging, achieving <5-minute mean time to detect (MTTD) for data quality failures.
- Extended data platform coverage to 3 additional business units by architecting new **ADF** ingestion pipelines connecting on-prem sources to **ADLS Gen2**.

### Trainee Data Engineer | PACECOM Technologies Pvt Ltd - Bengaluru

February 2023 - December 2023

- Built 6 end-to-end **ADF**-based ETL pipelines automating ingestion of structured datasets into **ADLS Gen2**-backed centralised repositories, replacing manual uploads and saving ~12 hours/week of analyst effort.
- Developed **Python** + **SQL** data processing jobs that standardised 8 disparate data sources into a unified schema, reducing ad-hoc data-prep requests to the engineering team by ~60%.
- Implemented automated data quality checks (null checks, referential integrity, schema validation) across ingestion pipelines, catching 95%+ of data anomalies before they reached reporting layers.
- Optimised **ADF** copy activities and **ADLS Gen2** data layouts (columnar **Parquet** + partitioning), reducing average load times by 28% for daily batch jobs.

## KEY PROJECTS

### Automated Data Flow Orchestration - Tech: Python | SQL | ADF | Databricks | ADLS Gen2 | Pandas

- Designed and delivered a fully automated **Azure**-native data platform ingesting from 5+ sources into a centralised **ADLS Gen2** lake, reducing manual intervention from daily to zero.
- Implemented modular **Python** + **SQL** transformation logic for data cleansing and standardisation, reducing downstream data defect rate by ~70%.
- Built **Azure Monitor** dashboards and alerting rules providing real-time pipeline health visibility; resolved recurring failure root causes, improving uptime from 94% to 99.2%.

### PySpark Pipeline Performance Optimization - Databricks - Tech: Python | PySpark | SQL | Databricks | ADLS Gen2 | Azure Monitor

- Identified and resolved processing bottlenecks in **PySpark** transformation jobs on **Databricks**, applying repartitioning and caching strategies that reduced average job runtime by ~40%.
- Rewrote 15+ complex **SQL** views and stored procedures - eliminated full table scans, cutting report generation time from 8 minutes to under 2 minutes.
- Implemented structured logging and **Azure Monitor** alerting across pipelines, reducing mean time to detect data quality failures to under 5 minutes.

### Microsoft Fabric Data Engineering - Tech: Microsoft Fabric | Lakehouse | OneLake | PySpark | Fabric Data Pipelines

- Built end-to-end **Lakehouse** pipelines in **Microsoft Fabric**, ingesting and transforming data from multiple sources into **OneLake** for unified analytics consumption.
- Developed **PySpark** notebooks within **Fabric Data Engineering** for cleansing and transforming large datasets, aligning schema design with downstream reporting needs.
- Configured **Fabric Data Pipelines** for scheduled and event-based ingestion, integrating with existing **ADF**-based workflows for a hybrid **Azure-Fabric** data architecture.
- Studied and mapped **Azure-to-Fabric migration** patterns - translating **ADF** pipelines and **Azure Synapse** workloads to **Fabric Data Pipelines** and **Lakehouse** equivalents, including OneLake storage consolidation considerations.

## EDUCATION

Bachelor of Engineering - PES College of Engineering, Mandya

## LANGUAGES

English (Professional) | Kannada (Native) | Hindi (Professional)